

Intro to Multiobjective Learning

0. Standard learning theory: single-objective risk minimization

$$\min_{h \in \mathcal{H}} R(h)$$

But, in practice, risk is multi-dimensional.

- generally, a single number is not enough to justify a business decision

Example. Suppose Uber wants to train a fleet of self-driving taxis

- safety
- profitability
- traffic avoidance
- legality
- fuel-efficiency
- safety during crashes ...

Example. Aligning LLMs to human values

Questions:

1. Algorithmic: how do we learn to minimize multiple risks?
2. Complexity-theoretic: how hard is multi-objective learning (MOL)?

Outline

1. Background from multi-objective optimization (MOO)
2. The learning setting
3. The multi-distribution learning problem
4. The multi-objective learning problem – unstructured formulation
5. The structured multi-objective learning problem

1. Background from multi-objective optimization

Let $R_1, \dots, R_K : \mathcal{H} \rightarrow \mathbb{R}$ be K risk functionals / objectives.

The goal:

$$\text{" minimize } (R_1(h), \dots, R_K(h)) \text{"}$$

⊗

• not a very well-defined goal. Why?

Usually, there is no clear winner. Instead, many incomparable trade-offs.

Example. Uber fleet policy: never leave parking spot.

- completely safe, but completely useless
- similar story in space of LLMs.

We probably prefer a more moderate trade-off.

Solution concept (classical formulation) — optimal trade-offs

A model $h^* \in \mathcal{H}$ is **Pareto optimal** in $\mathcal{H}$ if it is not possible to reduce one type of risk without worsening another: $\forall k \in [K]$ and $h \in \mathcal{H}$

$$R_k(h) < R_k(h^*) \implies \exists j \in [K] \text{ s.t. } R_j(h) > R_j(h^*).$$

Why Pareto optimality? At the minimum, our solution model should at least make an optimal trade-off — no way to strictly improve.

The reason ⊗ is not yet well-defined:

Do we return every Pareto optimal model?

Or, just one? In which case, which one?

↳ leads to two general approaches / algorithms in MOO.

(i) The a priori or non-adaptive method

A specific preference over trade-offs is supplied to the algorithm beforehand.

(ii) The a posteriori or adaptive method

The algorithm returns all trade-offs, and the user's preference may depend on the available options.

Solution concept (modern formulation)

Let S be a family of scalarizers $s: \mathbb{R}^k \rightarrow \mathbb{R}$.

A model $h_s \in \mathcal{H}$ is an s-trade-off solution if it minimizes:

$$h_s = \underset{h \in \mathcal{H}}{\operatorname{argmin}} \underbrace{s(R_1(h), \dots, R_k(h))}_{T_s(h) \leftarrow s\text{-trade-off functional}}$$

Example. Linear scalarization

$$S(v_1, \dots, v_k) = \lambda_1 v_1 + \dots + \lambda_k v_k \quad \text{where } \lambda_k \geq 0, \sum \lambda_k = 1.$$

Tchebycheff scalarization

$$S(v_1, \dots, v_k) = \max \{ \lambda_1 v_1, \dots, \lambda_k v_k \}$$

Fact. Under certain assumptions, solutions to these scalarizations are precisely the set of Pareto optimal solutions.

↳ the formulation via scalarization functions operationalizes Pareto optimality.

We can think of each $s \in S$ as a type of preference over trade-offs.

- in certain settings, we know the preference beforehand
- in others, we need to learn them.

2. The learning setting - supervised learning framework

$\mathcal{X}$ instance space

$\mathcal{Y}$ label space

$\mathcal{H}$ model class

$h: \mathcal{X} \rightarrow \mathcal{Y}$ model

$P_1, \dots, P_K$ data distribution over $\mathcal{X} \times \mathcal{Y}$

$\ell_1, \dots, \ell_K$ loss function $\ell(x, y, y') =$ loss incurred predicting y' on instance x with label y .

The specific risk functionals we consider:

$$R_k(h) := \mathbb{E}_{(x,y) \sim P_k} [\ell(x, y, h(x))].$$

The goal $\min_{h \in \mathcal{H}} (R_1(h), \dots, R_K(h))$ is not just an optimization problem

- we need to estimate R_k from data

• assume finite-sample access to P_k (and possibly to the marginal P_k^x).

Goal: for $s \in S$, use data from $P_1, \dots, P_K$ to minimize excess s -trade-off risk:

$$T_s(\hat{h}) - \min_{h \in \mathcal{H}} T_s(h) < \varepsilon.$$

Recall:

$$T_s(h) = s(R_1(h), \dots, R_K(h)).$$

Example. Self-driving car

- P_1 - data for whether an action will lead to a crash
- P_2 - data for fuel cost of an action
- P_3 - data about safety conditioned on crashing
- $\vdots$

Example. Fairness

For a collection of (possibly overlapping) groups $g_1, \dots, g_K$

let P_k = conditional distribution given instance belonging to g_k .

Note: in these examples, cannot just train a separate model for each objective.

3. The multi-distribution learning setting

- Haghtalab, Jordan, Zhou, "On demand sampling: learning optimally from multiple distributions" NeurIPS 2022.
- Aswathi, Haghtalab, Zhan, "The sample complexity of multidistribution learning for VC classes." COLT Open Problem 2023.
- Peng, "The sample complexity of multi-distribution learning". COLT 2024.
- Zhang, Zhan, Chen, Du, Lee. "Optimal multidistribution learning". COLT 2024.

This line of work belongs to the a priori or non-adaptive approach:

- care specifically about the minimax trade-off.

$$S(V_1, \dots, V_K) = \max \{V_1, \dots, V_K\}.$$

$$T_S(h) = \max_{k \in [K]} R_k(h).$$

Goal: minimize worst-case risk $\min_{h \in \mathcal{H}} \max_{k \in [K]} R_k(h).$

Setting: Given some budget n ,

- For $t=1, \dots, n$:
 - select $k \in [K]$ and receive a sample from P_k .
- Use collected data $(X_1, Y_1), \dots, (X_n, Y_n)$ to return $\hat{h} \in \mathcal{H}$
- Goal:

$$\Pr \left(T_S(\hat{h}) - \min_{h \in \mathcal{H}} T_S(h) \geq \varepsilon \right) < \delta.$$

$\uparrow$
 $S(v) = \max_{k \in [K]} v_k$

Results (Zhang et al. 2024).

$$n = O \left(\frac{VC(\mathcal{H}) + K}{\varepsilon^2} \right)$$

(Haghtakab et al 2022).

$$n = \Omega \left(\frac{VC(\mathcal{H}) + K}{\varepsilon^2} \right).$$

4. The multi-objective learning setting — unstructured formulation

- Cortes, Mohri, Gonzalo, Storchescu. "Agnostic learning with multiple objectives". NeurIPS 2020.
- Šukerík, Lampert. "Generalization in multi-objective machine learning" Neural Computing and Applications 2024.
- Wegel, Kovačević, Tifan, Yang. "Learning Pareto manifolds in high dimensions: how can regularization help?" AISTATS 2024.
- Wegel, So, Park, Yang. "On the sample complexity of semi-supervised multi-objective learning." Submitted 2025.

Goal: Given a family $\mathcal{S}$, use data to solve for all s -trade-offs:

$$\Pr \left(\forall s \in \mathcal{S}: T_s(\hat{h}_s) - \min_{h_s} T_s(h_s) \geq \varepsilon_s \right) < \delta.$$

↑
uniform guarantee over $\mathcal{S}$

" $\mathcal{S}$ -MOL" problem vs. " s -MOL" problem.

Result (Šukerík & Lampert):

$$N = O \left(\frac{VC(\mathcal{H}) \cdot K}{\varepsilon^2} \right)$$

Algorithm (MOL-ERM):

- For each $k \in [K]$, collect $n = \frac{V(\mathcal{H})}{\epsilon^2}$ data points from P_k

$$S_k = \{(X_k^{(1)}, Y_k^{(1)}), \dots, (X_k^{(n)}, Y_k^{(n)})\}$$

- Estimate the risk functional:

$$\hat{R}_k(h) := \frac{1}{|S_k|} \sum_{(x,y) \in S_k} \ell(x, y, h(x)).$$

- For each $s \in S$, return

$$\hat{h}_s := \operatorname{argmin}_{h \in \mathcal{H}} s(\hat{R}_1(h), \dots, \hat{R}_K(h)).$$

Result (Wegel et al 2025).

$$N = \Omega\left(\frac{V(\mathcal{H}) \cdot K}{\epsilon^2}\right)$$

Moreover, this lower bound holds in the worst-case even if

- 1) we know the Bayes optimal predictor for each P_k
- 2) we know the marginal distribution P_k^X over $\mathcal{X}$

↳ pessimistic result: making good trade-offs can be arbitrarily harder than minimizing individual risk even with strong priors.

Example. Goal: give movie recommendations to a group of K friends

- let $\mathcal{H}$ be the set of movies
- I can even tell you each person's favorite movie
- unfortunately, friend k 's favorite movie is completely abhorrent to friend $k+1$. So, in the worst case, this information is useless.

Question: How generalized is this worst-case setting? It is bad news if it is.

Consider the self-driving car policy

- if we need to do well across many objectives, we may need to ingest many features

→ but a feature important for one objective
(e.g. cost of fuel $\leftrightarrow$ profitability)

may be irrelevant to another (e.g. safety).

if model class $\mathcal{H}$ needs to be sufficiently rich to handle all S -tradeoffs,
this removes advantages from strong inductive biases / prior knowledge!

5. The structured multi-objective learning problem

a) Bias-variance trade-off in MOL:

Tension between statistical efficiency & Pareto optimality:

$$\min_{h \in \mathcal{H}} (R_1(h), \dots, R_K(h))$$

Bias-variance trade-off

- better trade-offs are possible over larger model classes $\mathcal{H}$
 - increased statistical costs of learning w/ larger $\mathcal{H}$
- in the single-objective setting, we often have good inductive biases or priors.
e.g. $\mathcal{G}$ a smaller model class that contains the Bayes model.

b) Augmenting the learning problem:

Assume:

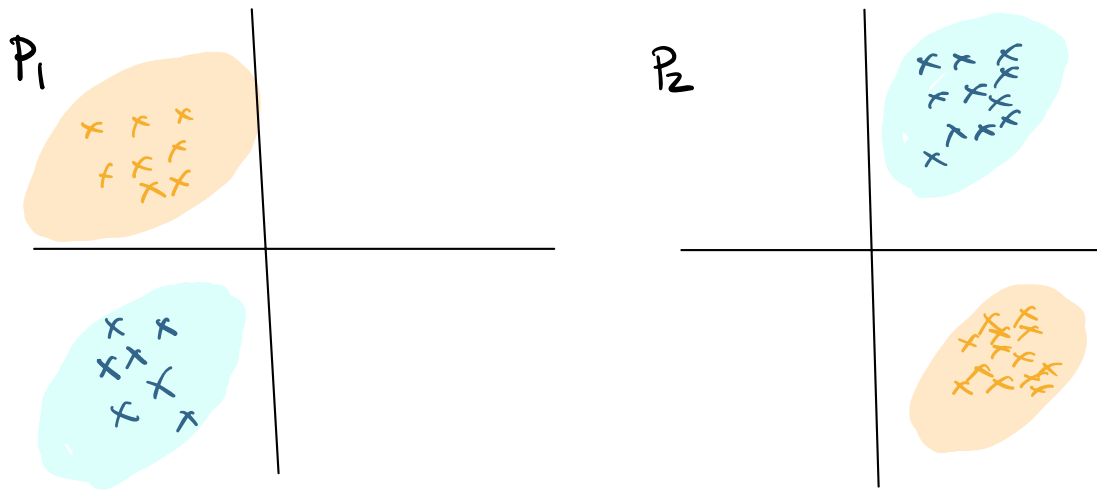
- for each $k \in [K]$, we have $\mathcal{G}_k \subset \mathcal{H}$ a model class guaranteed to achieve the same performance:

$$\min_{g \in \mathcal{G}} R_k(g) = \min_{h \in \mathcal{H}} R_k(h).$$

- $\mathcal{G}_k$ is less complex $\leftarrow$ models a good prior
(e.g. extract a good set of features for the k th task/risk).

Example. We know we can solve a set of risk minimization problems $R_1, \dots, R_K$ using $G_1, \dots, G_K$.

- but, to achieve good trade-off, need to consider larger model classes:



Here: linear classifier ^{G_K} perform great on either classification task
but if we want to perform well on both, need to consider
larger function classes _{$\mathcal{H}$}

c) Negative result, more formally

Proposition (Wegel et al 2025). Let S be the set of linear scalarization.

$$s(v) = \lambda_1 v_1 + \dots + \lambda_K v_K \quad (\text{convex combinations}).$$

Let $\mathcal{H}$ be a model class w/ $\forall \mathcal{H} \dim(\mathcal{H}) = d$ on X .

Let ℓ_K be the zero-one loss $\ell_K(x, y, y') = \mathbb{1}\{y \neq y'\}$.

For any collection of data distribution $(P_1, \dots, P_K)$, suppose

a learner is given n_K samples from P_K , a Bayes classifier $f_K^* : X \rightarrow Y$ and the marginal distribution P_K^X .

If the learner achieves:

$$\Pr \left(\forall s \in S, \quad T_S(\hat{h}_S) - \min_{h_S \in \mathcal{H}} T_S(h_S) < \varepsilon \right) \geq \frac{5}{6}$$

Then $n_1 + \dots + n_K \geq C \frac{d \cdot K}{\varepsilon^2}$ where C is some universal const.

- Thus, even if learner is told $G_K = \{f_K^*\}$
this does not help it solve S -ML.

d) Positive result

The issue: the zero-one loss is not informative.

Definition. Let Y be convex. A loss $\ell: Y \times Y \rightarrow \mathbb{R}$ is

a **Bregman loss** if there is a strictly convex, smooth potential

$$\phi: Y \rightarrow \mathbb{R} \quad \text{s.t.}$$

$$\ell(y, y') = \phi(y) - \phi(y') - \langle \nabla \phi(y), y - y' \rangle.$$

Examples.

- square-loss: $\ell(y, y') = \|y - y'\|^2$ $\phi(y) = y$
- Mahalanobis loss: $\ell(y, y') = \|Ay - Ay'\|^2$ $\phi(y) = Ay$
- cross-entropy loss / logistic loss:

$$\phi(y) = \sum y_i \log y_i$$

Important Property of Bregman losses

Lemma. If ℓ is Bregman, then:

$$1) \quad f^*(\cdot) = \underset{f}{\operatorname{argmin}} \quad R(f) = \underset{P}{\mathbb{E}}[Y | X = \cdot]$$
$$\underset{(x,y) \sim P}{\mathbb{E}}[\ell(x, y, f(x))]$$

2) Define the Bregman divergence

$$d(f, h) = \mathbb{E}_x [\ell(f(x), h(x))].$$

Then:

$$R(h) = R(f^*) + d(f^*, h).$$

This shows that the risk functional R is equal to $d(f^*, \cdot)$ up to a constant.

$\Rightarrow$ for informative losses such as Bregman losses, estimating the risk functional is equivalent to minimizing risk!

$\nearrow$ this is not true for zero-one loss

$$R(\cdot) = d(f^*, \cdot) + \text{constant}$$

Idea: estimate $R(\cdot)$ using $\hat{R}(\cdot) = d(\hat{f}, \cdot)$.

Lemma. Let ℓ be a Bregman loss, Lipschitz, and bounded.

Let f^* minimize R (ie. it is Bayes optimal).

If $R(\hat{f}) \leq R(f^*) + \epsilon$ then:

$$\|R - \hat{R}\|_{\infty} < c\sqrt{\epsilon}$$

where $\hat{R}(\cdot) = d(\hat{f}, \cdot)$.

In words: for informative / structured losses, if you can approximate the risk minimizer, then you can approximate the risk functional.

c.f. learning ideal points in a preference learning setting

Algorithm (Pseudo-labeling) — assume cheap access to P_k^x marginal

• For $k \in [K]$,

- get data from P_k
- run ERM over G_k to obtain $\hat{f}_k$
- use unlabeled data to construct $\hat{R}_k(\cdot)$

$$\hat{R}_k(h) = \hat{\mathbb{E}}_{X \sim P_k^x} \left[\ell(X, \hat{f}_k(X), h(X)) \right]$$

• For $s \in S$, solve:

$$\min_{h \in \mathcal{H}} s \left(\hat{R}_1(h), \dots, \hat{R}_K(h) \right).$$

Theorem (Wegd et al 2025).

Sample complexity of S-MOL for Bregman losses:

$$\frac{\text{VCdim}(G_1) + \dots + \text{VCdim}(G_K)}{\varepsilon^4} \quad \text{labeled data}$$

$$\frac{\text{VCdim}(\mathcal{H})}{\varepsilon^2} \quad \text{unlabeled data}$$

- data-dependent bounds are possible
- a localized generalization bound removes extra $\frac{1}{\varepsilon^2}$ factor in labeled sample complexity in certain settings.

6. Main takeaways

- 1) Scalarization formulation of MOO solution concept more amenable to algorithms & analysis than Pareto optimality.
- 2) There are two approaches to MOO: a priori / non-adaptive and a posteriori / adaptive preferences
- 3) Multi-objective learning should be formulated with more structure:
 - importance of the choice of loss
 - model classes for risk minimization v. trade-off optimization can be different

Thank You!

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